Alexander Hutchinson

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Education

University of Waterloo – BASc Systems Design Engineering, *Deans Honours*, GPA: **3.9** 2026

Skills

Languages: Java, Bash, Python, TypeScript, JavaScript, C#, C++, SQL, Dart, HTML & CSS
Frameworks & Libraries: Angular, NgRx, ASP.NET Core, Blazor, Flutter
Tools & Infrastructure: Git, Postman, ORDS, Oracle SQL Developer, MySQL, Redis, Jira, Confluence, Agile

Experience

GE Vernova Boston, USA
Software Engineer Intern June 2025 – Aug 2025

- Migrated 5 micro-applications from **Angular 2 to 13**, replacing legacy build systems with one Angular CLI pipeline, **unblocking feature work** and enabling up to **68% faster builds** via Angular 13's build cache
- Accelerated debugging workflows by utilizing **diagnostic logging** and GitHub Copilot to isolate complex dependency conflicts and resolve environment failures

RedIron Technologies London, Canada
Software Developer Intern Sept 2024 – Dec 2024

- Validated Oracle ORDS **REST endpoints** with **Postman** request/response tests, tracing failures to **PL/SQL queries** and fixing data errors in **15%** of endpoints
- Developed Count Sheet functionalities using C# and .NET, integrating front-end web development with **component-based architecture** to establish a proof of concept (POC)
- Programmed responsive Appearance settings using HTML, CSS, and **C# Blazor** components, enabling users to control the application's color scheme, font size, and component border radius

Tree of Life Canada Mississauga, Canada
Full Stack Developer Intern Jan 2024 – Apr 2024

- Developed **10+ components** on point-of-sales (POS) file management page with **Angular** & **Redux**, automating file upload, viewing, & validation, **saving 2+ developer hours** per week
- Wrote **API endpoints** utilizing **C# and .NET framework** to seamlessly fetch user information and upload files, serving as the backbone for the POS page
- Implemented email verification by leveraging **AWS Amplify** to automate email verification on user account creation enhancing user authentication process

Projects

Self-Hosted Homelab Aug 2025 – Present

- Architected a **two-node Ubuntu** homelab with a primary running **Docker** Compose services behind a nginx reverse proxy, and a secondary that receives encrypted restic backups so it can restore and take over if the primary fails
- Secured remote access with UFW, key-only SSH, and Tailscale, and built a **bubblewrap** sandbox that confines **AI agents** to a single project directory so they can run autonomously
- Deployed **Prometheus & Grafana** to monitor temperature and per-container CPU/memory, tracing a crash to thermal shutdown under sustained load

SpeakEasy Sept 2025 – Apr 2026

- Engineered a dynamic conversational interface in **Flutter** by implementing a model to synchronize character assets and real-time UI widgets with an asynchronous **state layer**
- Integrated backend pipeline using **FastAPI**, Celery, and Redis to process local audio recordings through an XGBoost disfluency classifier while offloading speech-to-text transcription to **AssemblyAI**

Game Automation Aug 2026 – Oct 2026

- Designed a two-pass pipeline that scanned **5,000+** Pokémon on an iPhone, reading stats via **OCR**, reconciling both passes in **SQLite**, and auto-tagging candidates in-game against configurable retention rules
- Built via **agentic programming** in a sandboxed environment, owning the architecture, retention rules, and validation while bridging the server to the phone through **Appium** on a Mac over an SSH tunnel